

Department of Telecommunications
Telecommunication Engineering Centre
(IoT Division)

F. No.- 1-3/2022-IoT/TEC

Dated 02.06.2022

Subject: Minutes of the first virtual meeting of National Working Group 20 (NWG-20) corresponding to ITU-T Study Group 20 on "Internet of Things (IoT) and Smart Cities & Communities (SC&C)" (reconstituted for the Study Period 2022-24), held on 23rd May 2022, 2:30 PM.

The first virtual meeting of the National Working Group 20 (NWG-20) corresponding to ITU-T Study Group 20, reconstituted for the Study Period 2022-24, was held on 23rd May 2022 at 2:30 PM through remote participation using C-DOT VC platform. The list of participants is available as *Annexure-1*.

1. Mr. Sushil Kumar, DDG (IoT) welcomed all the participants in the meeting. While briefing about the agenda of the meeting, he said that this is the first meeting of National Working Group 20 (NWG-20), reconstituted for the Study Period 2022-24, in line with ITU-T SG-20. He also mentioned that-

a. TEC is working in M2M/ IoT domain since 2014 and released 18 Technical Reports, with the outcomes intended to be used in the formulation of policies/ standards. Out of 18 TRs, five TRs as mentioned below, have been released in 2021-22:

1. IoT/ ICT Enablement in Smart Village & Agriculture
2. Code of practice for Securing Consumer IoT
3. Emerging Communication Technologies and use cases in IoT domain
4. IoT/ ICT Standards for Smart Cities
5. Framework of National Trust Centre for M2M/IoT Devices and Applications

TEC initiatives in M2M/ IoT domain have been listed in brief in the report on ***TEC Initiatives in M2M/ IoT Domain- An overview***. All the reports are available on TEC website (<https://www.tec.gov.in/M2M-IoT-technical-reports>).

b. Generally, ITU Study Groups have the study period of four years but due to Covid-19, previous study period which was from 2017-20, got extended upto 2022. The current study period is reduced to two years now i.e. from 2022 to 2024.

c. Based on the nominations received from industry, academia, association and Government, SDOs etc., this NWG has been reconstituted for the study period 2022-24.

- d. All the questions of ITU-T SG-20 which were there in previous study period, are the same for this study period also, as mentioned below-

Question	TITLE
Q1/ 20	Interoperability and interworking of IoT and SC&C applications and services
Q2/ 20	Requirements, capabilities and architectural frameworks across verticals enhanced by emerging digital technologies
Q3/ 20	IoT and SC&C architectures, protocols and QoS/QoE
Q4/ 20	Data analytics, sharing, processing and management, including big data aspects, of IoT and SC&C
Q5/ 20	Study of emerging digital technologies, terminology and definitions
Q6/ 20	Security, privacy, trust and identification for IoT and SC&C
Q7/ 20	Evaluation and assessment of Smart Sustainable Cities and Communities

- e. DoT is the state member in ITU from India, just like BIS is member in ISO/ IEC. TEC being the technical arm of DoT and the National Standardization body (NSB) for telecom and ICT Sector, is mandated to participate and submit contributions in the meetings of ITU and other standardization bodies such as APT, ETSI, 3GPP, oneM2M etc.
- f. ITU is having three sectors like Radio (ITU-R), Standardization (ITU-T) and the Development sector (ITU-D) and TEC mainly participates in Standardization sector.
- g. ITU-T SG-20 on "IoT and Smart Cities & Communities" was created by ITU in June 2015. On similar lines, National Working Group (NWG) – 20 was created in TEC in September 2015.
- h. The objective of the NWG-20 is to submit contributions in ITU-T SG-20, prepared through discussions in NWG-20 meetings. As a continuous process, the contributions are refined/ updated and presented in upcoming ITU-T SG -20 meetings. It takes around 2-3 years for getting them accepted as an international standard. Contributions may also be submitted in the form of proposed changes in the running/ existing contributions (temporary documents available on ITU website).
- i. **Next meeting of ITU-T SG-20 is scheduled on 18-28 July 2022** for which contributions are required to be prepared.
- j. Contribution related to IoT security may be prepared for the upcoming ITU-T SG-20 meeting. Members from C-DOT and USOF expressed their willingness to work on this topic. DDG (IoT) requested them to prepare some contribution (either new work item or on some existing work item in ITU).

- k. During the study period 2017-2020, two documents- **ITU-T Recommendation Y Suppl.53 (12/2018)** on IoT Use cases (having five IoT use cases from India and one from Egypt) and **ITU-T Recommendation Y Suppl.56 (12/2019)** on Smart City Use cases (having smart city use cases from Japan, Korea, UK and India), are having significant contributions from India.
 - l. ITU-T SG-20 is also having two focus groups FG-DPM (Focus Group on Data Processing and Management) and FG-AI4A (Focus Group on Artificial Intelligence (AI) and Internet of Things (IoT) for Digital Agriculture).
 - m. Recommendations released by ITU-T SG-20 are available on https://www.itu.int/ITU-T/recommendations/index_sg.aspx?sg=20.
 - n. Important ITU-T standards on IoT and Smart Cities have been listed in Annexure-1 of TEC Technical Report on **IoT/ ICT Standards for Smart Cities¹**.
 - o. It is expected to have larger participation from India in the ITU Study Group meetings by submitting more and more contributions.
 - p. DDG (IoT) requested members to express their views on how to move forward and their areas of interest.
2. Following members expressed their interest to contribute in following work areas:
 - a. Mr. Sharad Arora, Mashmari: Q2/ 20 and Q5/ 20
 - b. Mr. Sachin Chaudhari, IIIT Hyderabad: Q2/ 20 and Q5/ 20
 - c. Mr. Neelesh Mantri, BP: Q5/ 20 and Q7/ 20
 - d. Mr. Rajesh Gupta, USOF: IoT and 5G based applications in rural areas
 - e. Mr. Anubhav Prakash, SIA: will intimate later
 3. DDG (IoT) apprised members that TEC is currently working on work item Y. SRC **"Requirements for deployment of smart services in rural communities"** (copy attached). Members may provide their inputs on this contribution in track change mode only.
 4. On the request of few members, who were new to NWG-20, it was decided that a Newcomers' session on **"ITU-T SG-20 activities"** will be planned on 3rd June 2022 (Friday), for the new members in NWG-20 to familiarize them with the ITU-T study group working. Invitation has already been sent for this session.

¹ https://tec.gov.in/pdf/M2M/TR_IoT%20ICT%20Standards%20for%20Smart%20Cities.pdf

5. DDG (IoT) requested members to go through the recommendations released by ITU and give suggestions on areas where they can contribute.

In the end, DDG (IoT) thanked all the participants for attending the meeting.

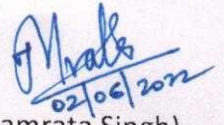
Way Forwards:

1. Next meeting will be held on 14th June 2022, 2:30 PM.

This is issued with the consent of DDG (IoT).

Encl: As above

Hindi version shall follow.



(Namrata Singh)

ADG(IoT)

TEC, New Delhi

Annexure-1

S. No.	Name	Organisation
1.	Sushil Kumar	TEC
2.	Ms. Ashima	TEC
3.	Ms. Namrata Singh	TEC
4.	Rajneesh Kumar	TEC
5.	Shekhar Singh	TEC
6.	Ms. Bhoomika Gaur	TEC
7.	Aurindam Bhattacharya	C-DOT
8.	Ravindra	Agent Tech
9.	Sharad Arora	Mashmari
10.	Deepesh Srivastava	Tejas Networks
11.	Dr. Punit Rathod	Qualcomm
12.	Desh Raj Bhadana	SIA- India
13.	Anubhav Prakash	SIA- India
14.	Sudipta	SIA- India
15.	Sudipta Bhaumik	Sterlite
16.	Sudipta Ghosh	WIPRO
17.	Sachin Chaudhari	IIITH
18.	Naveen Khichar	DoT
19.	N. Murali Krishna	DoT
20.	Neelesh Mantri	BP
21.	Ajay Mittal	TCS
22.	Kishor Narang	Narnix
23.	Ram	NCCS
24.	Abhijit Panicker	BIF
25.	Hrushikesh Reddy	Airtel
26.	Anshuman Roy	Sensorise
27.	Ms. Yashashri Shukla	USOF, DoT
28.	Rajesh Gupta	USOF, DoT
29.	Sahil Garg	USOF
30.	Shivanshu Tripathi	TEC
31.	Alok Mittal	STMicroelectronics
32.	Ms. Manisha Kumari	TEMA
33.	Manoj Desai	ARAI
34.	Vijay Madan	TSDSI